Day 73 – Network Monitoring & Management

Monitoring Checklist, Purpose, Commands, and Warning Signs

Overview

Day 73 focused on a different mindset than the rest of Phase 4.

Days 65-71 taught:

Build the network

Day 72 taught:

Fix the network

Day 73 taught:

Watch the network

The purpose is not to wait until users complain.

The purpose is to spot warning signs before the network breaks.

Monitoring Process

Whenever monitoring a network, follow this order:

1. Interfaces
2. Routing
3. Routing Protocols

4. Security Policies
5. DHCP
6. NAT
7. VLANs

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1. Interface Monitoring

Command

show ip interface brief

Where To Run It

Routers

(R0, R1, R2, Router3)

Purpose

Verify:

Interface state
Link health
Connected devices
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Healthy Output

up/up

Example:

G0/0 up up
G0/1 up up

Warning Signs

Interface Down

down/down

Possible causes:

Cable unplugged
Device powered off
Physical failure

Administratively Down

administratively down

Possible causes:

Shutdown command entered
Incorrect configuration

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2. Routing Monitoring

Command

show ip route

Where To Run It

Routers

Purpose

Verify the router knows:

Where networks exist
How to reach them

Healthy Output

Examples:

R 192.168.10.0
R 192.168.30.0
R 209.165.200.0

Warning Signs

Missing Route

Example:

Before:

R 209.165.200.0

After:

Missing

Possible causes:

RIP failure
Neighbor failure
Configuration change

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3. RIP Monitoring

Command

show ip protocols

Where To Run It

Routers running RIP

(R0, R1, R2)

Purpose

Verify:

RIP is running
Networks are advertised
Neighbors exist

Healthy Output

Routing for Networks:
10.0.0.0
192.168.20.0

Warning Signs

Missing RIP Network

Example:

Before:

10.0.0.0

After:

Missing

Possible causes:

Network statement removed
RIP misconfiguration

Exactly what happened in Day 72 Lab 2.

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4. ACL Monitoring

Command

show access-lists

Where To Run It

Router2

(or wherever ACLs exist)

Purpose

Verify:

ACL rules
Permit statements
Deny statements
Hit counters

Healthy Output

Example:

permit any
deny 192.168.10.0

with expected match counts.

Warning Signs

Huge Deny Matches

Example:

deny any
5000 matches

Possible causes:

Users being blocked
Bad ACL
Misconfigured security policy

Unexpected ACL

Example:

deny any

appears unexpectedly.

Possible cause:

Accidental security lockout
``

Exactly what happened in Day 72 Lab 3.

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5. DHCP Monitoring

Command

show ip dhcp binding

Where To Run It

DHCP Routers

Purpose

Verify:

Devices received addresses
DHCP is functioning

Healthy Output

Example:

192.168.10.2
192.168.20.2
192.168.30.2
``

Warning Signs

Missing Leases

Example:

No entries

Possible causes:

DHCP failure
DHCP pool issue
Client issue

Unexpected Lease Count

Example:

3 clients normally
20 clients suddenly

Possible causes:

New devices
Rogue device
Unauthorized users

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6. NAT Monitoring

Command

show ip nat translations

Where To Run It

Router2

Purpose

Verify:

Address translation
Internet connectivity

Healthy Output

Shows:

Inside Local
Inside Global

translations.

Warning Signs

Empty Translation Table

Example:

No entries

even while users are using the Internet.

Possible causes:

NAT failure
Routing failure
Traffic not reaching NAT router

Unexpected Translation Count

Example:

Thousands of translations

Possible causes:

Heavy traffic
Potential attack
Misconfiguration

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7. VLAN Monitoring

Command

show vlan brief

Where To Run It

Switches

Purpose

Verify:

VLANs exist
Ports belong to correct VLANs

Healthy Output

Example:

VLAN 10 HR
Fa0/1
VLAN 20 SALES
Fa0/2

Warning Signs

Missing VLAN

Example:

VLAN 10 gone

Possible causes:

Configuration loss
Deletion

Wrong Port Assignment

Example:

Fa0/1 moved to wrong VLAN

Possible causes:

Misconfiguration
Security issue

Day 73 Final Takeaway

The biggest lesson of Day 73 is:

Monitoring is looking for things
that have changed.

Ask yourself:

What does normal look like?

Once you know what's normal:

Missing routes
Interfaces down
ACL counters exploding
DHCP leases disappearing
Empty NAT tables
Wrong VLAN assignments

all become warning signs.

That's the transition from being a:

Network Builder

to a:

Network Operator

which is exactly what Day 73 was designed to teach before the Day 74 Enterprise Final Project. 🚀